

THE THREE CANONICAL MACHINE ARCHETYPES OF PHYSICAL AI

Distilling Physical Systems by Governing Conservation Laws, First-Binding Budgets, and Critical Failure Modes

1. MASS-DOMINATED

Mobility & Navigation



$$d_{stop} = v_0 \cdot \Delta t + v_0^2 / (2 \cdot a_{max})$$

- Exemplars:** AMRs, AVs, Drones, Mobile Bases
- First Budget:** Stopping Distance & Freshness (Δt)
- Failure Modes:** Kinetic collision; momentum breach
- Safety Gates:** MCU dynamic braking referee (1 kHz)

Delay converts directly to spatial drift (Δx)

2. CONTACT-DOMINATED

Manipulation & Assembly



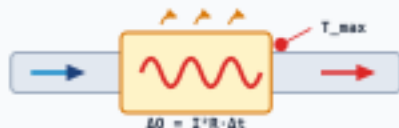
$$F_{contact} = k_c \cdot \Delta x, J_{ref} = N^2 \cdot J_{rotor}$$

- Exemplars:** Manipulators, Grippers, Humanoids
- First Budget:** Actuator Current & Gearbox Stress
- Failure Mode:** Gear tooth shear; workpiece damage
- Safety Gate:** Torque-derivative clamp & C^2 profiler

Position errors convert to massive forces (k_c)

3. FLOW-DOMINATED

Process, Thermal & Energy



$$dT/dt = (P_{in} - P_{out}) / C_{th}$$

- Exemplars:** Thermal loops, Inverters, Bioreactors
- First Budget:** Thermal Capacity & Phase Timing
- Failure Mode:** Thermal runaway; insulation blowout
- Safety Gate:** Hardware thermal observer & interlock

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